Startup Intelligence Challenge
AI-Powered Startup Investment Prediction

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Using NLP and machine learning to help investors identify promising startups with greater confidence, clearer signal, and lower risk.

The Problem: High Risk, Low Clarity

- Investors struggle to identify profitable startups early, when information is limited and decisions matter most.
- Early-stage investing carries high uncertainty, making it difficult to separate real potential from hype.
- Traditional decision-making lacks data-driven tools that can scale across many startup profiles.
- The cost of a wrong investment can be significant, creating a clear need for smarter prediction systems.

Our Solution: Predict Investment Potential

The proposed system evaluates startups using AI-powered analysis of text and structured attributes. By combining natural language processing with machine learning, it predicts whether a startup is worth investing in.

AI-based evaluation

Automates startup assessment using a repeatable, data-driven approach.

NLP + Machine Learning

Extracts meaningful signals from startup descriptions, categories, and taglines.

Clear output

Produces a simple investment label: **High Potential** or **Low Potential**.

Dataset Overview

The model is built on the **Pakistan Startup Census**, which provides a practical foundation for learning from real startup records.

Description — rich text data that captures what the startup does.
Category — helps place each company in its market context.
Founded Year — supports company age and maturity analysis.
Tagline — concise messaging that can reveal positioning and focus.

Data Preprocessing

Before modeling, the dataset was cleaned and standardized to reduce noise and improve learning quality. This stage turned raw startup entries into a more reliable input for prediction.

01

Missing values filled

Reduced gaps in the dataset so the model could learn from more complete records.

02

Founding year extracted

Structured time information to support company age analysis.

03

Company age created

Converted founding year into a more useful maturity signal.

04

Text cleaned

Removed stopwords, symbols, and other noise to improve NLP performance.

Feature Engineering

Feature engineering turned raw startup data into predictive signals that the model could use more effectively. The goal was to capture both text meaning and business context.

Description length as a proxy for clarity and detail.
Company age to reflect maturity and survival potential.
Keyword-based scoring for sectors such as AI, FinTech, and Health.
TF-IDF text features to represent important language patterns.
Category encoding to make startup types machine-readable.

Model Selection & Implementation

Two classification models were used to test the prediction task: Logistic Regression and Perceptron. The final choice favored Logistic Regression because it balanced performance and interpretability.

Logistic Regression

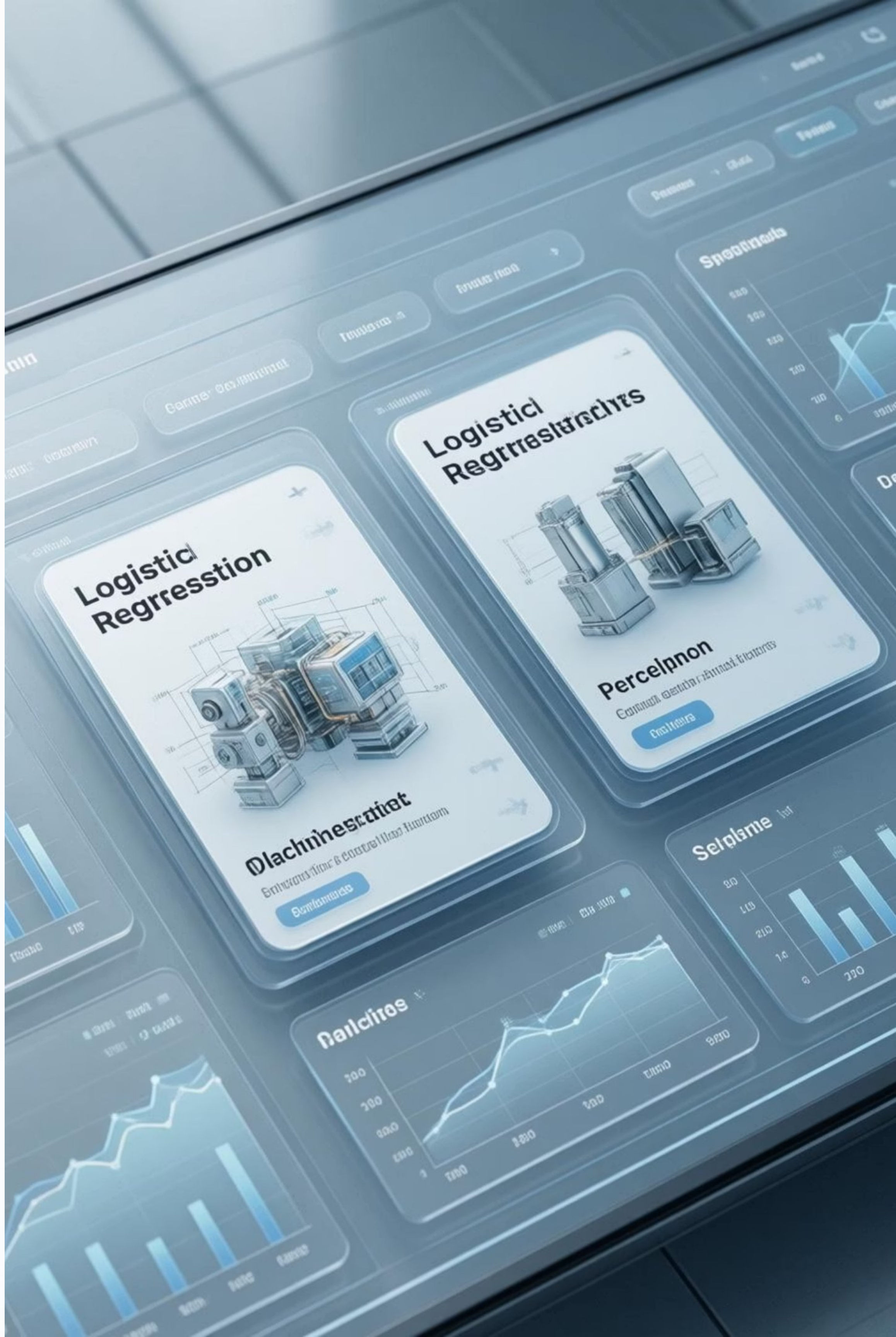
Selected for high accuracy, transparent behavior, and strong fit for binary classification.

Perceptron

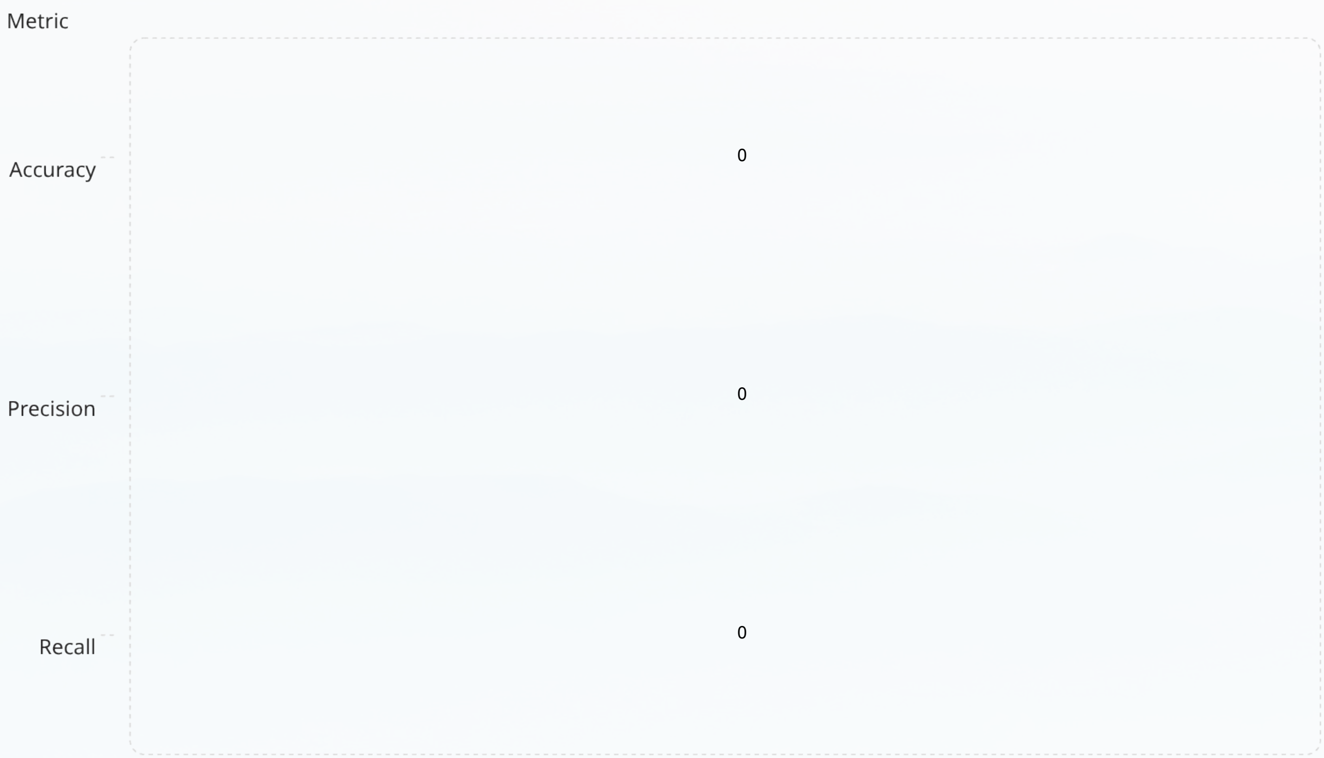
Used as a comparison model to evaluate baseline classification behavior.

Why it matters

In startup investing, interpretability is critical because stakeholders need to understand why a prediction was made.



Model Evaluation



The evaluation focused on the core classification metrics: accuracy, precision, and recall. Together, they show both overall correctness and how reliably the model identifies promising startups.

Confusion matrix insight: it shows correct versus incorrect predictions and helps reveal whether the model is trustworthy enough for investment decisions.

Insights & Feature Importance

The model surfaced several patterns that aligned with startup potential. These findings suggest that language, category, and maturity all contribute meaningfully to investment prediction.

AI, FinTech, Health

These startups tended to show stronger potential signals in the dataset.

Longer descriptions

More detailed descriptions often improved clarity and gave the model richer context.

Younger startups

Early-stage companies showed notable growth potential when supported by strong signals.

Failure Analysis & Conclusion

Failure analysis: some startups were misclassified because the text was noisy and keyword coverage was limited.
Improvements: future versions can use deeper models such as LSTM and BERT, along with more features.
Conclusion: AI can improve investment decisions, reduce risk, and scale to real-world startup screening.
Bottom line: a data-driven approach makes startup evaluation faster, more consistent, and more defensible.